LUCAS GONÇALVES

AI Engineer | LLM Agents · RAG · AI Guardrails & Evaluation

Marília-SP, Brazil (GMT-3) · Remote-ready · Open to relocation
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SUMMARY

AI Engineer with ~2.5 years building LLM agents, RAG and AI guardrails in Python (FastAPI, LangGraph, pgvector, GCP). Built a security gate that stopped all 21 severe vulnerabilities in a 22-project AI code-generation pilot, and co-led a fix that cut wrong-part returns from ~20% to 0% across 150+ dealers. Seeking AI Engineer roles in agentic and RAG systems, remote or with relocation.

KEY ACHIEVEMENTS

- Blocked 21 of 21 severe vulnerabilities before delivery in a 22-project pilot by building a deterministic security gate (Semgrep, Trivy) for a multi-agent code-generation platform.
- Demonstrated that an AI code-generation platform can validate a software idea for ~R$3K in under 2 weeks, instead of ~R$200K over ~6 months, by instrumenting the cost of every run.
- Calibrated a research RAG agent to accept 96.7% of in-domain questions while letting only 3.3% of off-topic ones through, using an ROC analysis over 120 labeled questions.

EXPERIENCE

Unipac (Grupo Jacto) | Data & AI Assistant | Pompeia-SP (Hybrid, part-time) | 09/2026 – Present
Packaging and polymers manufacturer of the Jacto Group, building an internal multi-agent platform where business users describe an idea and AI agents build, test and security-check the application before a developer approves it.

- Replaced the model's self-reported security verdict with a deterministic gate (Semgrep with 20 rules, Trivy) running in a network-less sandbox and covered by 154 tests; it blocked all 21 severe findings across a 22-project pilot before any code was delivered.
- Re-architected agent orchestration from an LLM-driven loop, where the model itself chose which agent ran next, into a deterministic four-wave scheduler running up to 12 agent phases (LangGraph, LangChain, FastAPI, GCP Cloud Run); 12 of 14 end-to-end runs were approved.
- Automated the repair of failed checks by routing each finding back to the agent that wrote the code, resolving 14 of 21 gate failures across 13 runs without human input.
- Mitigated runaway agent behavior after a single model response fired 2,179 identical tool calls, adding a 9-layer guardrail stack (call caps, loop breaker, circuit breaker) so failures stop safely instead of burning budget.
- Instrumented per-run cost across 13 end-to-end runs (US$1.01–6.96, median US$2.63 and 17 minutes), showing a software idea can be validated for ~R$3K in under 2 weeks instead of ~R$200K over ~6 months.
- Documented metadata for 15+ previously undocumented tables in a ~3.8 TB, ~25-billion-row BigQuery lakehouse and contributed to its data-quality standards, so AI agents query company data with the right context.

PM Accelerator | AI Engineer | Remote (Miami, FL, USA) | 06/2026 – 08/2026
U.S. AI product accelerator; built an AI college-admissions advisor with an international team of 8 contributors (Cohort 9).

- Led the product launch as backend AI engineer, owning the path from a student's question to a grounded answer: retrieval, prompt assembly, token-budgeted memory, routing between a fast and a stronger model, and a 90-second turn limit (Python, FastAPI, PostgreSQL).
- Designed hybrid retrieval that ranks exact school-name and alias matches ahead of semantic search (pgvector HNSW) across ~1,800 U.S. institutions, answering "not enough data" instead of guessing from a similar school.
- Eliminated a class of hallucinations in code rather than in the prompt: unanswerable questions are refused before any model call, and a verifier flags any rate, cost or GPA that the retrieved data does not support.
- Secured each student's data with ownership checks on every query, enforced by a test harness that fails the build on any cross-student leak.
- Reduced the CI install footprint ~9x (~800 MB to ~90 MB) by serving embeddings through ONNX instead of PyTorch.
- Expanded the backend suite from 539 to 561 passing tests in the final week, closing grounding and ordering edge cases before launch.

Jacto | After-Sales – Data & Automation | Pompeia-SP (Hybrid, part-time) | 11/2024 – 06/2026
Global agricultural machinery manufacturer with 5,000+ employees and presence in 100+ countries.

- Co-led the fix of wrong-part returns from ~20% (2025) to under 5% (2026) and 0% in 06/2026, against a 6% Six Sigma target, writing the return manual approved across 150+ dealers; the fix avoided an estimated ~R$100–135K in freight in 1H/2026.
- Audited a 13,000+ part return flow (~R$6.5M per semester); the findings fed the manual, which corrected more than half of the discrepancies.
- Automated the collection of address data from 99 regional pages of a foreign registry (Python, Selenium, pandas), translating it from Russian and exporting it ready for Salesforce, saving up to 5 hours of manual work per day.
- Streamlined the parts-return system with its developers, validating 20+ improvements, including a sync across SAP, Salesforce, physical counts and the platform that cut record loading from 5–10 minutes to under 1 minute.
- Interpreted English–Portuguese at up to 3 international live events, supporting communication between foreign visitors and local teams.

PROJECTS

SmartB100 – Agentic RAG for soil fertility | Research project (FAPESP) | github.com/LukeSantossz/sb100_agents

- Soil-fertility questions need answers grounded in agronomy, and off-topic questions must be refused. Designed the agent architecture and calibrated its domain filter with an ROC sweep over 120 labeled questions, accepting 96.7% of in-domain questions at a 3.3% false-positive rate; 370 tests, 90% line coverage and 15 architecture decision records (Python, FastAPI, Qdrant, Ollama, LangGraph).

VisioSoil – Offline soil-texture classifier | Undergraduate thesis | github.com/LukeSantossz/visiosoil-app

- Soil texture analysis usually requires a lab. Built a Flutter app that classifies texture from a photo fully offline, computing 26 image descriptors on the phone, verified by 992 Dart and 810 Python tests in an 8-job CI; won 3rd place at FETEPS 2025 (1,300+ entries) and was presented at ICPA 2026 (Flutter/Dart, scikit-learn).

tweet-sentiment-analysis – Emotion classification | Personal project | github.com/LukeSantossz/tweet-sentiment-analysis

- Fine-tuned RoBERTa for emotion classification, raising macro F1 from 0.584 to 0.887 (+52%) over the baseline on the 2,000-row test split, and rewrote preprocessing in Rust to run 4.1x faster on 1M tweets (PyTorch, Hugging Face, Rust).

my-framework – Engineering quality gates | Open source (MIT) | github.com/LukeSantossz/my-framework

- Shipped v0.8.0 with 7 deterministic quality gates and binaries for 5 platforms, adopted across 5 of my repositories.

SKILLS

Languages: Python, SQL, TypeScript, Go
Frameworks & AI: FastAPI, LangGraph, LangChain, PyTorch, Hugging Face Transformers, scikit-learn, RAG, LLM evaluation
Cloud & Data: GCP (Cloud Run, BigQuery), PostgreSQL/pgvector, Qdrant, Docker
Tools: GitHub Actions, Semgrep, Trivy, Ollama, OpenAI API
Soft Skills: International remote collaboration, team leadership, technical writing, English–Portuguese interpretation

EDUCATION & CERTIFICATIONS

- Technologist Degree in Big Data for Agribusiness, Fatec Shunji Nishimura, Brazil | 01/2024 – 11/2026 (expected)
- Certified AI Engineer, PM Accelerator (2026) · Google Foundations of Cybersecurity (08/2026) · Meta Introduction to Android Development (01/2026)
- Poster: "Soil Texture Classification by Image: Deep Feature Learning vs. Handcrafted Methods for Precision Agriculture", 17th ICPA (07/2026)
- Awards: 3rd place, 16th FETEPS 2025 (1,300+ entries); 1st place, Bootcamp Jovem Programador Unimar (2023) · Volunteer researcher, FAPESP 2024/00985-1
- Languages: Portuguese (native), English (professional working proficiency)